Phase 1: SiWx917 Wi-Fi + MQTT on FreeRTOS

Milestone plan and status. Updated 2026-09-23. Part of the larger Matter-fabric project (SiWx917 = Wi-Fi endpoint, MG240 = Thread sensor, Orange Pi Zero 3 = Matter hub).

1. Hardware

- Radio board: SiWX917-RB4338A - the SiWG917 SoC (Cortex-M4 application core + ThreadArch core that runs the wireless stack), Wi-Fi 6 + BLE 5.4, 8 MB flash.
- Mainboard: Si-MB4002A (BRD4002A) Wireless Pro Kit - carries the J-Link programmer and USB-C power. The board plugs into the mainboard; this is the official Silicon Labs SoC development setup (SiWx917-PK6031A = mainboard + this radio board).
- Sensors available: SI7021 (I2C, fixed address 0x40) is the Phase 1 sensor; AM2320 (I2C 0x5C) and DHT12 (I2C 0x5C / DHT single-wire) are spares.
- Note: the SiWx917 is a Silicon Labs SoC. The Cypress/Infineon CYW43389 is NOT part of this system.

2. Software stack

- SDK: Simplicity SDK v2026.6.1 (github.com/SiliconLabs/simplicity_sdk) installed via the Silicon Labs installer (Studio or CLI), plus the WiSeConnect v4.x extension - a Simplicity Studio EXTENSION that adds SiWx91x Wi-Fi/BLE support to that SDK. It is NOT a standalone SDK, and there is no separate 'classic' SDK.
- Application core: FreeRTOS on the M4 core; the connectivity stack runs on the ThreadArch core.
- Verified for v4.1.1: the 'Wi-Fi - Embedded MQTT Client' example is available in SoC mode - it connects to a broker and publishes + subscribes. This is the starting point.
- MQTT broker: mosquitto, first on the PC, later on the Orange Pi Zero 3 (which also runs the Phase 2 Matter stack).

3. Tasks

1. W0 - Hardware check: BRD4338A on the WPK, USB-C power. The out-of-box demo should start BLE advertising (device name BLE_CONFIGURATOR); commission it to Wi-Fi with the Simplicity Connect app. Confirms board + J-Link before any development.
2. W1 - Toolchain: install Simplicity SDK v2026.6.1 + the WiSeConnect v4.x extension; BRD4338A target appears in EXAMPLE PROJECTS AND DEMOS when the board is connected.
3. W2 - Create the 'Wi-Fi - Embedded MQTT Client (SoC)' example for BRD4338A; set SSID/PSK; point the broker at a local mosquitto; build; flash via the WPK J-Link; verify publish + subscribe from the console. (This is the MVP.)
4. W3 - Own the app: replace the demo payload with periodic JSON telemetry; keep the message-building logic in the portable sensor_pub module (see section 5).
5. W4 - CI: headless slc build of the project in GitHub Actions + host unit tests for the portable module. (SBC deploy CI arrives with Phase 2.)

4. What this milestone covers on the skills list

- RTOS (Tier 1, item 1): FreeRTOS on the M4 application core, W1-W3.
- MQTT pipeline (Teton-mirroring): device -> broker starts here; the SBC ingest completes it in Phase 2.
- CI on real hardware: starts at W4 (headless firmware build + host unit tests).
- I2C from the datasheet (Tier 1, item 3): comes with the SI7021 integration in W3.

5. Portable sensor_pub module (built and tested 2026-09-23)

The W3 payload module is already written and unit-tested on the host (no board required), so W3 on the device is integration, not development: src/sensor_pub.h, src/sensor_pub.c, test/test_sensor_pub.c in the sensor-pub folder.

- Pure C11, no heap, no threads: identical behavior on the M4 core (FreeRTOS) and in the CI host build.
- Fixed buffer (160 bytes), static storage in the hot path - fits the no-heap discipline.
- Transport is a function pointer: the firmware wraps the SDK MQTT client; the host test uses a mock that records every byte.
- JSON contract (stable, agreed with the SBC ingest): {"dev":"<id>","ts":<uptime_s>,"temp":21.5,"rh":55.0,"bat":3300}. Out-of-range temp/RH become null so the SBC can drop the frame.

Verification (host, gcc 13, -Wall -Wextra -Werror): build OK, 21 assertions across 14 check groups, ALL PASSED. Covered: exact payload contract, range boundaries (-40/85 degC, 0/100 %RH), truncation -> error, device-id validation, mock transport byte pass-through, error-code propagation.

6. Open items / watch-outs

6. Confirm the WiSeConnect SDK version at install time and that the Embedded MQTT Client example lists SoC mode for BRD4338A (checked in v4.1.1 docs; verify against the downloaded release).
7. SI7021 I2C on the BRD4338A: check the radio board schematic for the free I2C pins and whether the board's own sensors occupy the bus (W0 is the moment to verify this physically).
8. MQTT over TLS: the embedded client example supports plain MQTT first; TLS adds a certificate step - do plain MQTT to the local broker in W2, TLS later.

7. Handover to Phase 2

When W0-W4 are done, the same Simplicity SDK (2026.6.1 + WiSeConnect extension, SoC mode) is used to flash the Matter-over-Wi-Fi firmware for the fabric; the SiWx917 is then the Wi-Fi endpoint alongside the MG240 Thread sensor, with the Orange Pi Zero 3 running the border router, matterjs-server and the matter-mqtt-bridge.